Cloud Computing systems
Lecture 3

Service models of cloud computing

IaaS, PaaS

dr hab. inż. Aneta Poniszewska-Marańda, prof. PŁ
Instytut Informatyki Politechniki Łódzkiej

Cloud computing

Definition

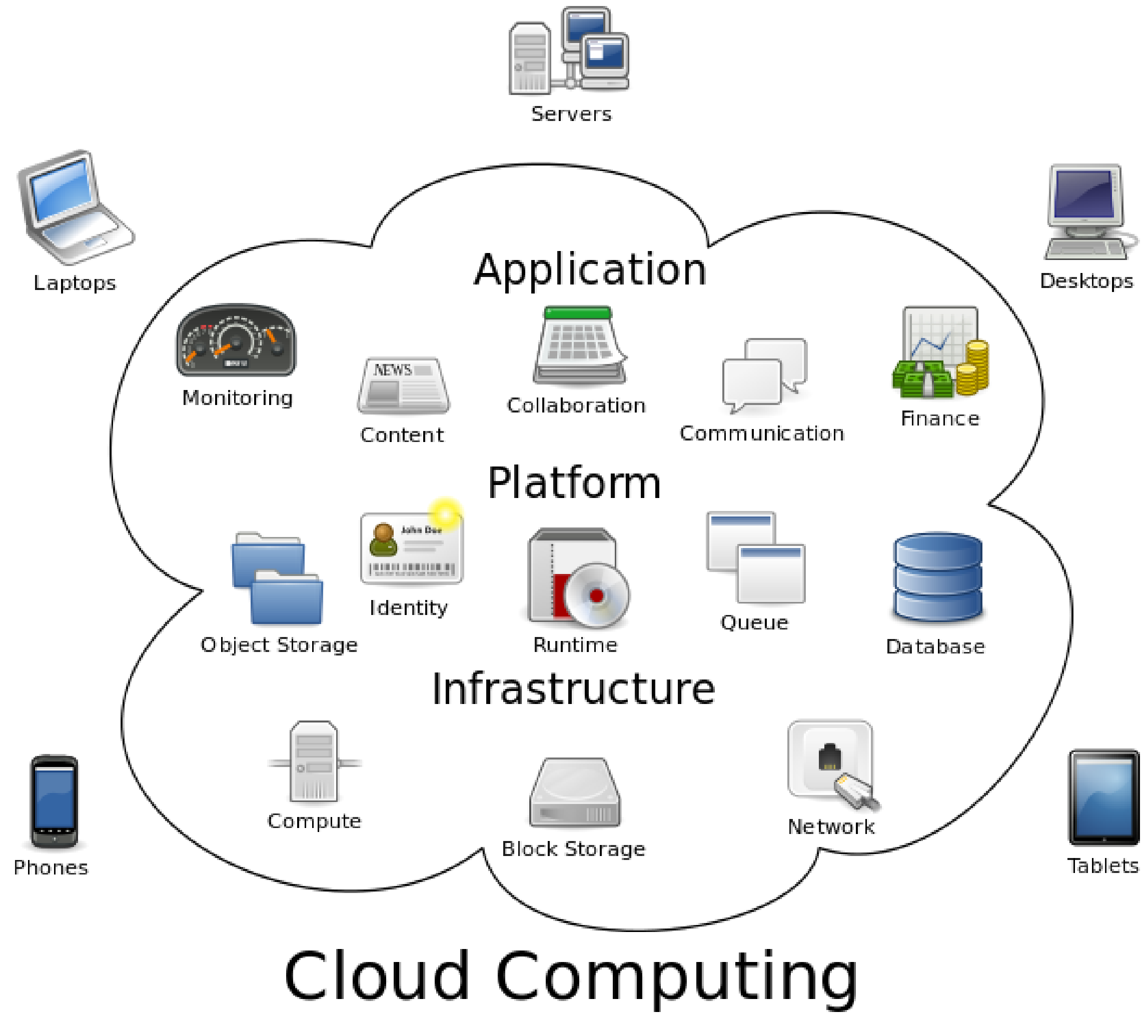
- National Institute of Standards and Technology (NIST) provides the following definition of cloud computing:

*"**Cloud computing** is a model for enabling ubiquitous, convenient, on-demand network access to a shared pool of configurable computing resources (e.g. networks, servers, storage, applications, and services) that can be rapidly provisioned and released with minimal management effort or service provider interaction".*

dr hab. inż. Aneta Poniszewska-Marańda
Instytut Informatyki Politechniki Łódzkiej

Cloud computing

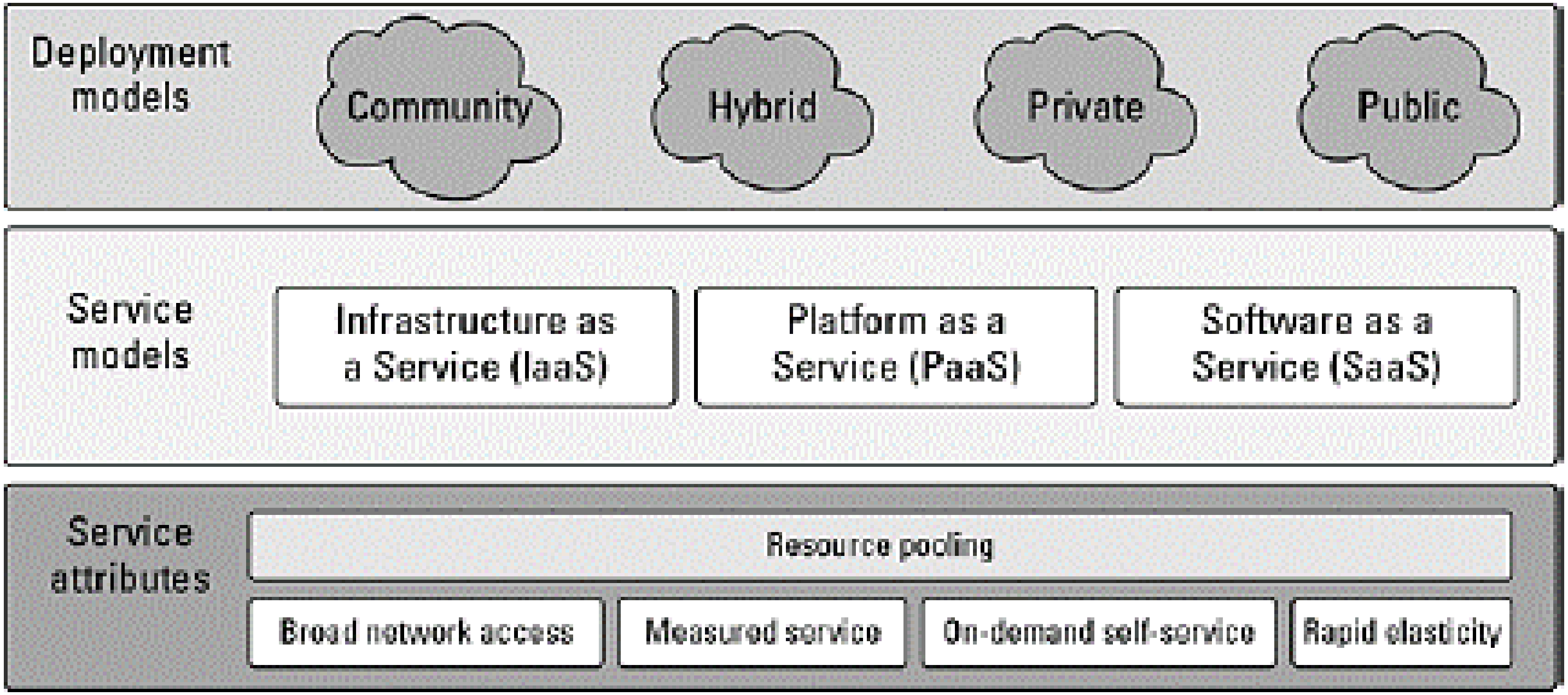
The general structure of cloud computing



dr hab. inż. Aneta Poniszewska-Marańda
Instytut Informatyki Politechniki Łódzkiej

Cloud computing

- in order to provide a precise definition of cloud computing the NIST introduced the division of cloud computing into **deployment models** and **service models**



dr hab. inż. Aneta Poniszewska-Marańda
Instytut Informatyki Politechniki Łódzkiej

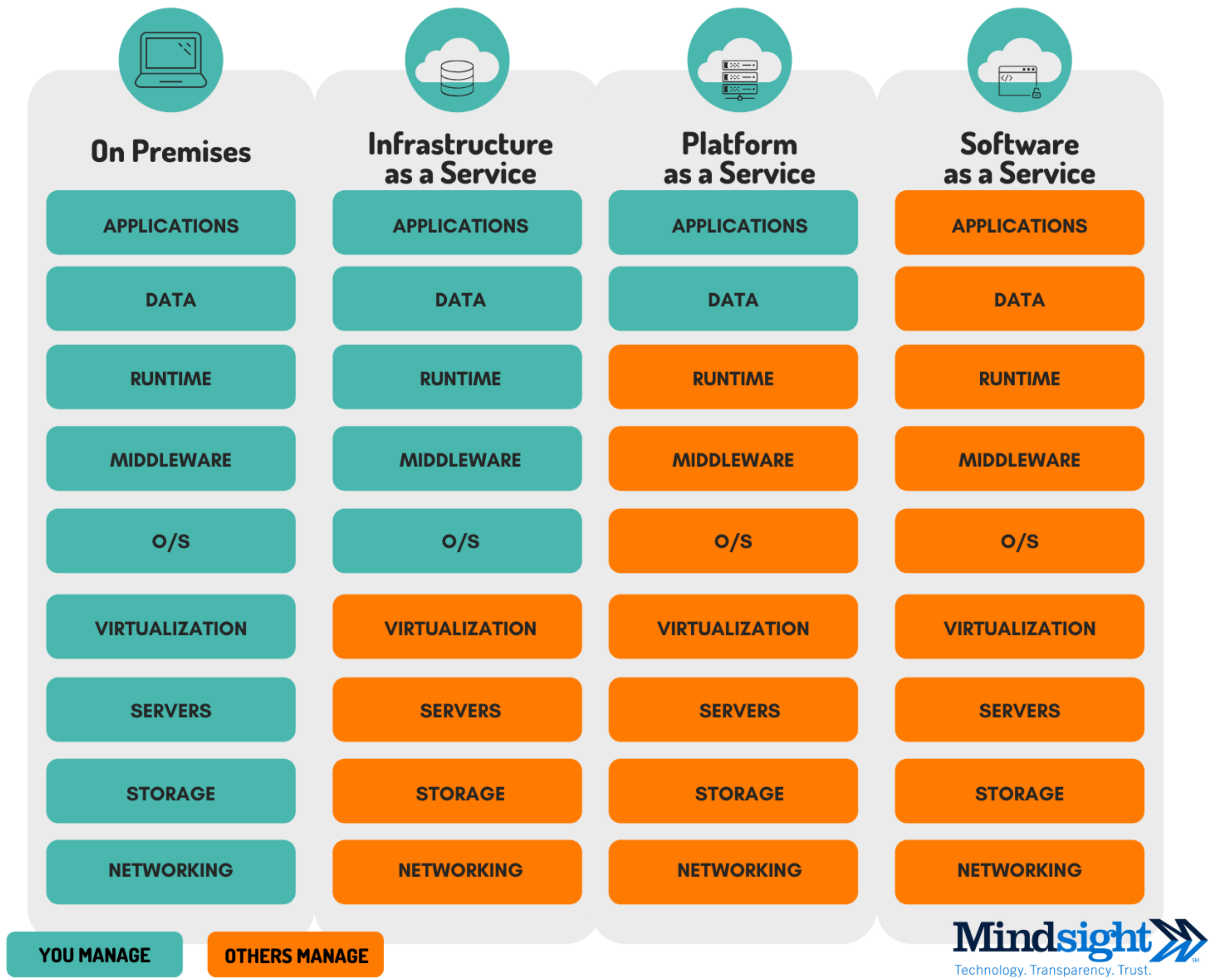
Service models of cloud computing

- nowadays, more and more new functions are implemented in the cloud model
- it is a matter of time to reach the peak of virtualization – transferring all the software (including the operating system) to the server, and the user installing a thin client with only communication interfaces
- most cloud services fall into one of four general categories:
  - **infrastructure as a service (IaaS)**
  - **platform as a service (PaaS)**
  - **software as a service (SaaS)**
  - **serverless**
- these categories are sometimes called cloud computing stock because they extend each other's capabilities

dr hab. inż. Aneta Poniszewska-Marańda, prof. PŁ
Instytut Informatyki Politechniki Łódzkiej

Service models of cloud computing

Division of cloud computing by services



Service models of cloud computing

Division of cloud computing by services

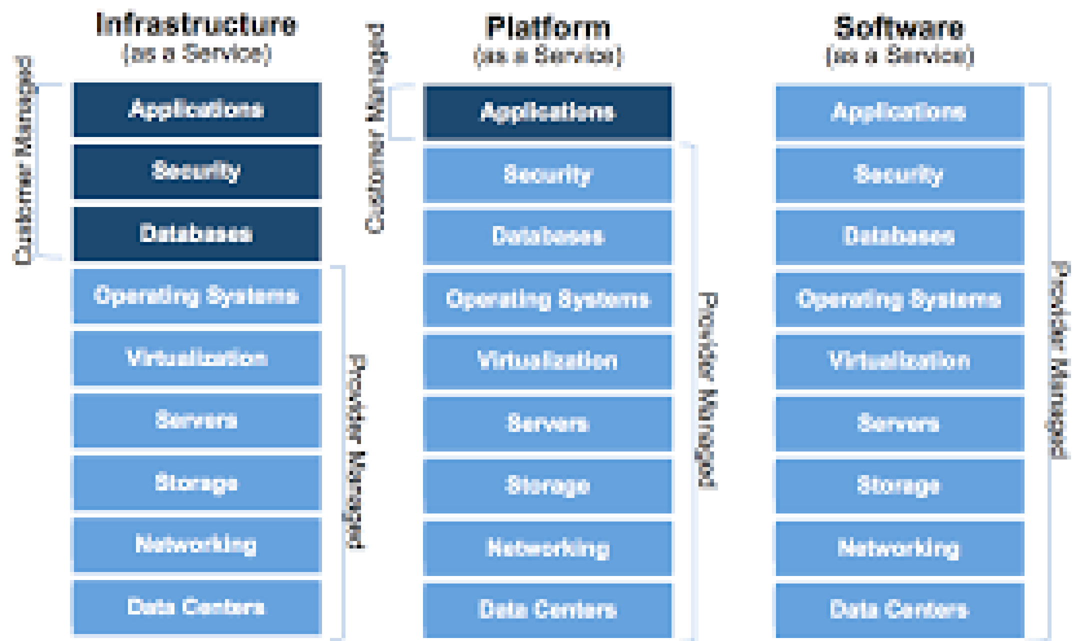
Chmura w 5 odmianach

Typ chmury	Fizycznie	Dla	Co
S+S	+ Aplikacje lokalne	Specjalista IT	Intune
SaaS	Aplikacje	Użytkownik	BPOS, CRM
PaaS	OS + Platforma	Programista	Azure
IaaS	Sprzęt (serwery)	Specjalista IT	Azure VM*
Kolokacja	Datacenter	Specjalista IT	-

dr hab. inż. Aneta Poniszewska-Marańda, prof. PŁ
Instytut Informatyki Politechniki Łódzkiej

Service models of cloud computing

Division of cloud computing by services



dr hab. inż. Aneta Poniszewska-Marańda, prof. PŁ
Instytut Informatyki Politechniki Łódzkiej

Service models of cloud computing

IaaS Cloud – Infrastructure as a service

- ***IaaS – Infrastructure as a service*** is a type of cloud computing in which a third-party provider hosts virtualized computing resources over the Internet

- in an IaaS model, a third-party provider hosts hardware, software, servers, storage and other infrastructure components on behalf of its users
- IaaS providers also host users' applications and handle tasks including system maintenance, backup and resiliency planning

dr hab. inż. Aneta Poniszewska-Marańda, prof. PŁ
Instytut Informatyki Politechniki Łódzkiej

Service models of cloud computing

IaaS Cloud – Infrastructure as a service

IaaS is the utilization of APIs to manage the lowest levels of network infrastructure, including networking, storage, servers, and virtualization

- **Common Use Cases:**
  - IaaS is the most flexible service model for cloud computing – it is especially effective for start-ups and organizations looking for agile scaling
  - also preferred by businesses that seek greater control over their resources

dr hab. inż. Aneta Poniszewska-Marańda, prof. PŁ
Instytut Informatyki Politechniki Łódzkiej

Service models of cloud computing

IaaS Cloud – Infrastructure as a service

- the lowest available function level in which the virtual machine images provided by the service provider will have the computing power we have defined
- it can be e.g. a virtual machine with a given processor power, amount of RAM and disk space supported by the Windows Server operating system
- another option of the IaaS service may be the indicated amount of disk space, e.g. for storing data backup

Service models of cloud computing

IaaS Cloud

- IaaS platforms offer highly scalable resources that can be adjusted on-demand
- -> this makes IaaS well-suited for workloads that are temporary, experimental or change unexpectedly
- other characteristics of IaaS environments include:
  - automation of administrative tasks
  - dynamic scaling
  - desktop virtualization
  - and policy-based services

dr hab. inż. Aneta Poniszewska-Marańda, prof. PŁ
Instytut Informatyki Politechniki Łódzkiej

Service models of cloud computing

IaaS Cloud

- customers of IaaS model pay on a per-use basis, typically by the hour, week or month
- some providers also charge customers based on the amount of virtual machine space they use
- this *pay-as-you-go* model eliminates the capital expense of deploying in-house hardware and software
- however, users should monitor their IaaS environments closely to avoid being charged for unauthorized services

dr hab. inż. Aneta Poniszewska-Marańda, prof. PŁ
Instytut Informatyki Politechniki Łódzkiej

Service models of cloud computing

IaaS Cloud

- IaaS providers own the infrastructure -> systems management and monitoring may become more difficult for users
- also, if an IaaS provider experiences downtime, users' workloads may be affected
  - for example, if a business is developing a new software product, it might be more cost-effective to host and test the application through an IaaS provider
  - once the new software is tested and refined, it can be removed from the IaaS environment for a more traditional in-house deployment or to save money or free the resources for other projects

Service models of cloud computing

IaaS Cloud

- leading IaaS providers include *Amazon Web Services (AWS)*, *Windows Azure*, *Google Compute Engine*, *Rackspace Open Cloud*, and *IBM Smart Cloud Enterprise*, *Digital Ocean*

dr hab. inż. Aneta Poniszewska-Marańda, prof. PŁ
Instytut Informatyki Politechniki Łódzkiej

Chmura IaaS

Typical IaaS applications in enterprises (1)

- **Software testing and development.** Teams can quickly configure and remove test and development environments, accelerating the introduction of new applications to the market. Scaling up and down development and test environments with IaaS is fast and economical.
- **Hosting websites.** Sharing websites with IaaS can be cheaper than using traditional web hosting.

dr hab. inż. Aneta Poniszewska-Marańda, prof. PŁ
Instytut Informatyki Politechniki Łódzkiej

Chmura IaaS

Typical IaaS applications in enterprises (2)

- **Storage, backup and restore.** Organizations avoid capital expenditure on storage and warehouse management work, which usually requires qualified personnel to manage data while meeting legal requirements and complying with standards. The IaaS solution is useful for handling sudden and unforeseen as well as ever-increasing demand for storage. It can also facilitate the planning and management of backup systems.
- **Web applications.** The IaaS solution provides all the infrastructure necessary to support Web applications, including storage, Web servers, application servers and network resources. Using IaaS, organizations can quickly deploy Web applications and easily scale infrastructure up and down when demand for applications is unpredictable.

dr hab. inż. Aneta Poniszewska-Marańda, prof. PŁ
Instytut Informatyki Politechniki Łódzkiej

Chmura IaaS

Typical IaaS applications in enterprises (3)

- **High performance calculations.** High-performance computations performed with supercomputers, computer networks or computer clusters help solve complex problems involving millions of variables or calculations. Examples of applications are earthquake and protein folding simulations, weather and climate forecasts, financial modelling, and product design assessment.
- **Big data analysis.** Big data is a popular term for huge data sets potentially containing valuable patterns, trends and connections. Searching data to find these hidden patterns requires enormous computing power, and the IaaS solution provides this power in an economical way.

dr hab. inż. Aneta Poniszewska-Marańda, prof. PŁ
Instytut Informatyki Politechniki Łódzkiej

Chmura IaaS

Advantages of IaaS solution (1)

- **Eliminates capital expenditure and reduces current costs.** The IaaS solution avoids the initial expenses associated with setting up and managing a local data centre, making it an economical option for start-ups and enterprises testing new ideas.
- **Facilitates business continuity and disaster recovery.** Ensuring high availability, business continuity and disaster recovery is expensive because it requires many technologies and a large number of employees. However, after concluding the appropriate service level agreement (SLA), the IaaS solution can reduce this cost and provide normal access to applications and data in the event of failure or shutdown.

dr hab. inż. Aneta Poniszewska-Marańda, prof. PŁ
Instytut Informatyki Politechniki Łódzkiej

Chmura IaaS

Advantages of IaaS solution (2)

- **Rapid Innovation.** After deciding to launch a new product or launch a new initiative, the necessary computing infrastructure can be ready in a few minutes or hours, not days or weeks, and sometimes even months that would be necessary to prepare it in the company.
- **Faster response to changing business conditions.** The IaaS solution allows you to quickly increase the amount of resources to handle a sudden increase in demand (e.g. during holidays), and then reduce it during periods of less activity to reduce costs.
- **Allows you to focus on the most important activities of the company.** Thanks to the IaaS solution, the team does not have to deal with IT infrastructure and can focus on the most important activities of the organization.

dr hab. inż. Aneta Poniszewska-Marańda, prof. PŁ
Instytut Informatyki Politechniki Łódzkiej

Chmura IaaS

Advantages of IaaS solution (3)

- **Increased stability, reliability and serviceability.** If you use the IaaS solution, you do not need to maintain or upgrade software and hardware, or troubleshoot hardware problems. After concluding the relevant contract, the service provider ensures infrastructure reliability and ensures its operation in accordance with the SLA.
- **Greater security.** After concluding the appropriate service contract, the cloud service provider protects your applications and data using security that may be better than obtainable locally.
- **Provides users with faster access to new applications.** Because you do not need to configure the infrastructure before developing and delivering the application, users will be able to access it faster with the IaaS solution.

dr hab. inż. Aneta Poniszewska-Marańda, prof. PŁ
Instytut Informatyki Politechniki Łódzkiej

Service models of cloud computing

PaaS Cloud – Platform as a service

- *PaaS – Platform as a service* is a category of *cloud computing services* that provides a *platform*
- allowing customers to develop, run and manage *Web applications* without the complexity of building and maintaining the infrastructure typically associated with developing and launching an application

dr hab. inż. Aneta Poniszewska-Marańda, prof. PŁ
Instytut Informatyki Politechniki Łódzkiej

Service models of cloud computing

PaaS Cloud – Platform as a service

PaaS offers users the capability to build or deploy applications using tools (i.e. programming languages, libraries, services) without maintaining the underlying infrastructure – users instead have control over the applications themselves

- **Common Use Cases:**
  - PaaS is highly available and highly scalable
  - it gives organizations the ability to build and create new services and solutions without the need for highly skilled developers focused on software maintenance
  - PaaS is preferred by IT in hybrid cloud environments

dr hab. inż. Aneta Poniszewska-Marańda, prof. PŁ
Instytut Informatyki Politechniki Łódzkiej

Service models of cloud computing

PaaS Cloud – Platform as a service

- ***Platform as a service (PaaS)*** is a complete development environment and implementation environment in the cloud including resources enabling the delivery of any solution, from simple cloud-based applications to complex applications for enterprises using the cloud
- needed resources bought from the cloud service provider using the payment model according to actual use
- access to them is obtained through a secure internet connection

dr hab. inż. Aneta Poniszewska-Marańda, prof. PŁ
Instytut Informatyki Politechniki Łódzkiej

Service models of cloud computing

PaaS Cloud

- PaaS solution includes infrastructure (servers, storage and network), as well as middleware, development tools, business analysis services, database management systems, etc.
- PaaS solution is designed to support the entire life cycle of a web application: creation, testing, implementation, management and updating
- PaaS solution helps to avoid expenses and work related to the purchase of software licenses, basic application infrastructure and middleware, container orchestrators or development tools and other resources
- client manages developed applications and services, and the cloud service provider usually does everything else

Service models of cloud computing

PaaS Cloud

- PaaS can be delivered in two ways:
  - as public cloud service from a provider, where the consumer controls software deployment and configuration settings, and the provider provides the networks, servers, storage and other services to host the consumer's application
  - as software installed in private data centres or public infrastructure as a service and managed by internal IT departments
- two primary programming languages for PaaS are Java and.NET

dr hab. inż. Aneta Poniszewska-Marańda, prof. PŁ
Instytut Informatyki Politechniki Łódzkiej

Service models of cloud computing

PaaS Cloud

- PaaS model provides an environment for developers and companies to create, host and deploy applications, saving developers from the complexities of the infrastructure side (setting up, configuring and managing elements such as servers and databases)
- PaaS can improve the speed of developing an application, and allow the consumer to focus on the application itself
- with PaaS, the consumer manages applications and data, while the provider (in public PaaS) or IT department (in private PaaS) manages *runtime*, *middleware*, *operating system*, *virtualization*, *servers*, *storage and networking*

dr hab. inż. Aneta Poniszewska-Marańda, prof. PŁ
Instytut Informatyki Politechniki Łódzkiej

Service models of cloud computing

PaaS Cloud

- development tools provided by the vendor are customized according to the needs of the user
- user can choose to maintain the software, or have the vendor maintain it

dr hab. inż. Aneta Poniszewska-Marańda, prof. PŁ
Instytut Informatyki Politechniki Łódzkiej

Service models of cloud computing

PaaS Cloud

- PaaS offerings may also include facilities for application design, application development, testing and deployment, as well as services such as
  - team collaboration
  - web service integration and marshalling
  - database integration, security
  - scalability, storage, persistence
  - state management, application versioning
  - application instrumentation
  - and developer community facilitation
- besides the service engineering aspects, PaaS offerings include mechanisms for service management, such as monitoring, workflow management, discovery and reservation

dr hab. inż. Aneta Poniszewska-Marańda, prof. PŁ
Instytut Informatyki Politechniki Łódzkiej

Service models of cloud computing

PaaS Cloud

- **advantages** to PaaS are primarily that it allows for higher-level programming with dramatically reduced complexity
  - overall development of the application can be more effective, as it has built-in infrastructure
  - maintenance and enhancement of the application is easier
  - also be useful in situations where multiple developers are working on a single project involving parties who are not located nearby

dr hab. inż. Aneta Poniszewska-Marańda, prof. PŁ
Instytut Informatyki Politechniki Łódzkiej

Service models of cloud computing

PaaS Cloud

- primary **disadvantage** would be the possibility of being locked in to a certain platform
  - however, most PaaSes are relatively lock-in free
- other possible disadvantage include the relative youth of the cloud service model, the lack of support for.NET by many providers
- and that the value and definition of PaaS has been at times misunderstood by those working in IT

dr hab. inż. Aneta Poniszewska-Marańda, prof. PŁ
Instytut Informatyki Politechniki Łódzkiej

Service models of cloud computing

PaaS cloud – types: public, private and hybrids

- PaaS *public* model is derived from the SaaS model and is positioned between the SaaS and IaaS models
- *private* PaaS cloud can be downloaded and installed on the company's infrastructure or in the public cloud
  - after installing the software on one or several computers, private PaaS organizes applications and databases into one parent platform
- PaaS *mobile* cloud provides development opportunities for programmers and designers of mobile applications
- *open* PaaS cloud based on free software (*open source software*) does not include hosting services, but provides the possibility of using *open source software* with solutions that allow cloud providers to run applications in this environment

dr hab. inż. Aneta Poniszewska-Marańda, prof. PŁ
Instytut Informatyki Politechniki Łódzkiej

Service models of cloud computing

PaaS Cloud

- PaaS model can be found in the following types of systems:

1. *Programs supporting software development*
   - programs adapted to existing SaaS applications force suppliers and PaaS users to purchase a SaaS application subscription
2. Autonomous environments
   - autonomous environments of the PaaS model are not technically, licensed or financially dependent on SaaS applications or websites
   - are designed to provide a general software development environment
3. ……

dr hab. inż. Aneta Poniszewska-Marańda, prof. PŁ
Instytut Informatyki Politechniki Łódzkiej

Service models of cloud computing

PaaS Cloud

- PaaS model can be found in the following types of systems :

3. *Exclusive application delivery environments*
   - PaaS „only delivery” service usually refers to management services such as security and scalability on demand
   - the service does not include software development, troubleshooting and testing
   - it is possible to provide such services offline (e.g. via the Eclipse plugin)

There are many types of PaaS providers. They all offer application management and deployment environment, along with other integrated services.

dr hab. inż. Aneta Poniszewska-Marańda, prof. PŁ
Instytut Informatyki Politechniki Łódzkiej

Service models of cloud computing

PaaS Cloud

- example PaaS providers include *Salesforce*, *AWS Elastic Beanstalk*, *Heroku*, *Google App Engine (GAE)*, *OpenShift*, *Microsoft Azure*, *Apache Stratos*, *IBM Cloud*

PaaS cloud

Typical applications of PaaS in organizations (1)

- **Development environment.** The PaaS solution provides an environment that developers can extend to create or customize cloud-based applications. The PaaS solution allows developers to create applications using embedded software components in a similar way to creating macros in Excel. Cloud features such as scalability, high availability and multi-availability are included by default, which reduces the amount of work that developers need to do.
- **Additional services.** PaaS solution providers may offer other services that increase application capabilities, such as workflow, catalogue, security and planning.

PaaS cloud

Typical applications of PaaS in organizations (2)

- **Analytics or business analysis.** The tools provided as services as part of the PaaS solution enable organizations to analyse and search data to find information and patterns and predict results for better forecasting, increasing return on investment, as well as making better decisions on product designs and other business decisions.

PaaS cloud

Advantages of PaaS solution (1)

- because PaaS provides infrastructure as a service, using it *has the same advantages as using IaaS*
- its additional features - middleware, development tools and other business tools - provide additional benefits:
- **Shortening coding time.** PaaS development tools can reduce the time it takes to create new applications thanks to pre-created application components built into the platform, such as workflow, directory services, security features, search, etc.
- **Additional programming possibilities without hiring new employees.** Platform components as a service can provide the development team with new opportunities without having to hire employees with the necessary skills.

dr hab. inż. Aneta Poniszewska-Marańda, prof. PŁ
Instytut Informatyki Politechniki Łódzkiej

PaaS cloud

Advantages of PaaS solution (2)

- **Easier creation for many platforms, including mobile ones.** Some service providers provide programming capabilities for many platforms, such as computers, mobile devices and browsers, so you can create cross-platform applications faster and easier.
- **The use of advanced tools at a low cost.** Thanks to the real-time payment model, individuals and organizations can use advanced development software and advanced analytics and business analysis tools that they could not buy for financial reasons.

dr hab. inż. Aneta Poniszewska-Marańda, prof. PŁ
Instytut Informatyki Politechniki Łódzkiej

PaaS cloud

Advantages of PaaS solution (3)

- **Support for geographically distributed development teams.** Because access to the development environment is accessed via the Internet, development teams can collaborate on projects even when team members are in distant locations.
- **Effective application life cycle management.** The PaaS solution provides all the functions necessary to operate a web application throughout its entire life cycle (creating, testing, implementing, managing and updating) within the same integrated environment.